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Examining the nonlinear and threshold effects of the 5Ds built environment to land values using interpretable machine learning models

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Abstract: Previous studies have extensively explored the critical influence of the built environment on land values, but the non-linear relationship has yet to be fully revealed. This study aims to uncover the non-linear relationship between land values and the five built environment dimensions using machine learning algorithms and Shapley Additive exPlanation (SHAP). The results highlight that the Gradient Boost Decision Tree (GBDT) outperforms eXtreme Gradient Boosting (XGBoost), Ordinary Least Squares (OLS), and Multiscale Geographically Weighted Regression (MGWR) in land value estimation, exhibiting higher R^2 and lower Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). The results illustrate that density and destination accessibility are the most significant contributors, contributing 32.18% and 31.94% to land value variation, respectively. We observed that the top three factors affecting land values are the built-floor area ratio, the number of floors and the number of restaurants. Additionally, the results revealed the non-linear relationship between the built environment and land values, suggesting that maintaining built environment features at optimal thresholds may increase land values. Neglecting interaction effects may lead to bias in determining relationships between land values and the built environment. This study contributes to the literature by providing non-linear and threshold identification evidence in land value determinants, offering valuable insights for urban planners and real estate managers.

Keywords: the built environment, land values, housing price, GBDT, SHAP, non-linear relationships, machine learning.

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1 Introduction

Land is a limited resource, unique in terms of location. Owning land and a house can provide happiness and is considered a dream of middle and low-income workers. Homeownership was central to the American perception of a successful life (Adams, 1984, Rohe et al., 2013). However, the soaring costs of land and houses in central areas can pose challenges, leading to financial strain and loan repayment burdens and often requiring buyers to cut back on other expenses (Chen et al., 2023).

Prospective buyers conscientiously evaluate land values and housing prices, considering various determinants before purchasing. Scholars have extensively examined the factors influencing land values (Mulley and Tsai, 2017, Doan and Tran, 2020, Song and Hao, 2022). Research in this domain has shed light on how different variables impact land affordability, contributing to effective real estate market management. Real estate characteristics, transportation accessibility, and the surrounding environment are crucial determinants of land values. For instance, utilizing a hedonic price model, Diao and Ferreira (2010) delved into the influence of the built environment on residential property values, underscoring the significance of transit accessibility and job opportunities. Guo et al. (2016) recommended that urban planners consider factors such as land use mix, entertainment locations, and public transport accessibility in street design, while investors should focus on property features, physical conditions, and point of interest locations. The impact of the built environment on land values and housing prices has been a focal point in studies by Zhong and Li (2016), Bagheri and Shaykh-Baygloo (2021). These studies collectively emphasize the intricate interplay between the built environment and the economic aspects of real estate, providing valuable insights for both urban planning and investment considerations.

While significant strides have been made in understanding the impact of various factors on land values, there exists a need for more focused investigations into the nuanced mechanisms through which the built environment influences these values. Firstly, the effect of the built environment on house prices and land values remains inconsistent (Aurand, 2010), introducing noteworthy research gaps. For instance, there are mixed results on the significant effects of transportation accessibility on house prices and land values. While most studies explored the positive impact of railway and metro stations on house prices (Li et al., 2019b, Tan et al., 2019), contradictory results are evident in the findings of Berawi et al. (2020) and Zhong and Li (2016), challenging the conventional wisdom. Li et al. (2019a) introduce an additional layer of complexity, revealing that while public and private service amenities hold significance in the inner-city housing market, transportation accessibility emerges as the most sought-after factor for potential buyers. Secondly, a critical observation arises from the reliance on linear assumptions in previous studies investigating the influence of various factors on house prices (Gibbs et al., 2018, Cao et al., 2019). For example, Kong et al. (2007) employed the OLS model to identify the effects of urban green space on house prices and revealed a

straightforward positive association. Kim et al. (2007) explored how highway traffic noise impacts land prices, demonstrating a linear relationship: a 1% increase in noise resulted in a 1.3% decrease in land price. Diao and Ferreira (2010) demonstrated that proximity to the transit station contributed to a 2.37% increase in property prices for one standard deviation increase. The mixed results mentioned above underscore the need to explore non-linear associations and thresholds, as conventional models employing linear assumptions may fall short of capturing the true complexity of the relationships. Linear assumptions may lead to overestimation or underestimation of the impact of dependent variables, especially considering the variations in coefficients across different ranges of variable values. Therefore, addressing these gaps by examining non-linear relationships, threshold effects, and global and local interactions is imperative for advancing our understanding in this field.

Recently, non-linear machine learning algorithms, such as gradient-boosting decision trees, support vector machines, extreme trees, and convolutional neural networks, have gained popularity due to their ability to overcome the limitations of conventional methods, resulting in significant improvements in prediction accuracy (Liu et al., 2021a). However, these machine-learning models are often considered black-box, raising concerns about interpretability due to their complex structures (Gao et al., 2019). As an interpretability tool, Shapley addresses this limitation by ranking the relative importance of variables, identifying non-linear relationships, and determining thresholds (Zhou et al., 2022). GBDT and XGBoost, in particular, have been recognized for their effectiveness in quantifying impacts with significant prediction accuracy, as demonstrated in studies by Hao et al. (2022), and Doan et al. (2024).

To address the aforementioned knowledge gaps, this study employed GBDT and XGBoost and compared their performances in predicting land values to determine the most accurate algorithm. The selected best model is then combined with Shapley to identify the relative importance and non-linear relationships between the built environment and land values. Understanding the non-linear relationship between the built environment and land values can contribute to urban planning, real estate investment strategies, and policy recommendations. Planners and policymakers may use threshold effects to develop more flexible zoning regulations that account for the varying impacts of different built environment features on land values. Additionally, based on insights into non-linear relationships, policy regulations can be developed to support sustainable urban development, balancing economic considerations with those of the built environment. Specifically, this study aims to answer the following questions:

- (1) Which dimension of the 5D built environment is the most significant in influencing land values?
- (2) What are the specific mechanisms through which the built environment influences land values?

(3) When does the influence of built environment features on land values shift from positive to negative and vice versa?

The remainder of this article is organized as follows. Section 2 provides a comprehensive overview of the relationship between the built environment and residential property prices, as well as methods used for property price prediction. Section 3 presents the research area, data sources, analytical framework, and methodology. Section 4 presents the model results and highlights non-linear relationships between built environment features and land values, followed by the discussion and conclusion.

2 Literature review

2.1 The built environment and property price

Scholars have established linear relationships between built environment factors and property prices, focusing on destination accessibility, design, and transit distance.

Destination accessibility has been identified as a significant factor influencing property prices. Land values decreased with increasing distance to the city centre due to reduced accessibility (Alonso, 1964). Enhanced accessibility to urban facilities, such as shopping malls (Zhang et al., 2020a), public parks (Espey and Owusu-Edusei, 2001), urban lakes (Zhang et al., 2020b), schools (Livy, 2017, Hu et al., 2020), hospitals and sports centers (Glaesener and Caruso, 2015), film production studios (Chang, 2023), and green areas (Łaszkiewicz, 2023). Zahirovic-Herbert and Gibler (2021) positively influence house prices. Zhang et al. (2020a) observed a positive impact on prices from large high-end and leisure shopping malls in non-core areas. Doan (2023) noted a negative relationship between distance to hospitals and industries, showing a decrease in land values with increasing distance. Using a spatial economic model, Wen et al. (2014) found a positive association between education facilities and house prices, reporting a 2% to 5.4% increase in house prices with improved primary and secondary school quality.

Furthermore, design characteristics play a vital role in determining house prices by influencing buyers' decisions. Existing literature extensively examines the impact of house features on housing prices. Residential areas with large lots and gardens, providing ample space for recreational activities, have been associated with higher housing prices (Marcus and Sarkissian, 1986). Kang et al. (2021) discovered that smaller house areas may have a higher potential for appreciation, aligning with the study conducted by Doan (2023), who identified a negative association between land size, frontage, and land values. Moreover, Reusens et al. (2023) developed a novel framework to assess the impact of dwelling characteristics on the house price index. On a macro-scale, the urban environment significantly influences housing prices, with factors such as urban appearance (Jia et al., 2022), urban growth (Bagheri and Shaykh-Baygloo, 2021), and the surrounding urban context (Chen et al., 2022), socio-economic (Li et al., 2022). For instance, (Jia et al., 2022)

highlighted the importance of street view quality, emerging as the most crucial factor affecting house prices in areas with highly educated residents. In another study focusing on house price data in Hangzhou, Wen and Tao (2015) emphasized the substantial impact of polycentric urban design on house prices, with the price elasticity increasing over an 8-years.

On the other hand, the distance to transit is a crucial factor in housing price studies, and various scholars have delved into how transportation systems influence house prices (Liu et al., 2021b, Morikawa et al., 2023). However, the results of these studies have been mixed and, at times, conflicting. Most studies found a positive influence of transport accessibility on house prices (Yang et al., 2020, McCartney and Rosenvasser, 2023). For instance, Garrett (2004) observed a 32% increase in residential land prices by constructing an urban railway nearby. As the number of transportation services increases, accessibility rises, leading to reduced transportation costs and increased land prices. Debrezion et al. (2010) found a close relationship between railway accessibility, distance to railway stations, railway service quality, and house prices. Similarly, Yang et al. (2023) evaluated the impact of walking and cycling accessibility on house prices, revealing a heterogeneous influence of transportation accessibility within a city. However, some researchers reported negligible or negative impacts of transportation accessibility on house prices. Wagner et al. (2017) indicated an 8% price reduction for properties within 1.5 km of rail transit, while Zhong and Li (2016) found a negative impact of rail transit on single-family property prices. Furthermore, Liu et al. (2022) discovered a mixed effect (initial increase and subsequent decline) in the impact of bus accessibility on house prices, consistent with the findings of Cohen and Brown (2017). Beyond transportation, scholars have also explored the impact of other factors on housing prices, including environmental issues (Zhang et al., 2022), socio-economic (Mirkatouli et al., 2018, Doan, 2021), and noise (Zheng et al., 2020).

2.2 Methods for property price prediction

Previous studies have employed diverse methodologies for property price prediction, encompassing both traditional and modern approaches. While conventional methods often lean towards hedonic modelling, recent years have witnessed an increased application of machine learning algorithms.

Hedonic modelling is a primary method to illuminate the relationship between independent and dependent variables, effectively uncovering associations between property prices and their features. Among the various hedonic models utilized in property price studies, Ordinary Least Squares (OLS) and Geographically Weighted Regression (GWR) models have been prominent (Wu et al., 2014, Thompson et al., 2023). OLS models, characterized by straightforward interpretation, are widely employed, while GWR models factor in spatial variation based on geographical location, enhancing their adaptability (Huang et al., 2017). These models were used to determine the relationship between house prices and traffic noise (Blanco and Flindell, 2011), land use intensity

(Kang, 2017), and urban greening (Dell'Anna et al., 2022). Despite the good interpretability of these statistical models, they are constrained by their assumption of linear relationships between house prices and determinants. This limitation may overlook potential non-linear relationships and thresholds, emphasizing the need for more adaptive methodologies.

Scholars have increasingly turned to machine learning algorithms in property price studies in recent years, highlighting their advantages over conventional statistical methods (Zurada et al., 2011). Researchers employed machine learning algorithms such as random forest, support vector machines, artificial neural network (ANN), gradient-boosting decision trees and extreme gradient boosting. For instance, Gu et al. (2011) combined genetic and support vector machines in house price prediction. Wang et al. (2014) utilized support vector machines and particle swarm optimization for price prediction, demonstrating robust performance in real estate value prediction. In comparative studies, researchers have assessed the performance of different machine learning models to identify the most suitable algorithms for specific datasets. Yilmazer and Kocaman (2020) highlighted the superior performance of the random forest algorithm over stepwise linear regression in house price prediction. Abidoye and Chan (2017) emphasized the efficacy of ANN for housing price prediction in Nigeria, establishing it as a reliable method. Chen et al. (2016) developed a housing price prediction model by ranking the performance of six machine learning models to predict rental prices. Ma et al. (2020) compared the performance of six machine learning models and found that random forest was the most suitable model. Comparative performance studies of machine learning algorithms were also conducted by Hu et al. (2020), and Chen et al. (2023). Despite the prominence in performance prediction, machine learning algorithms are often considered black boxes, making them difficult to interpret. Therefore, combining the advantages of machine learning with interpretation techniques is necessary to explore the non-linear relationship between land values and other urban objectives, an aspect currently lacking in the literature.

In summary, previous studies have employed conventional and modern methods to investigate the relationships between the physical urban environment and land values. However, these studies often analyze pairwise relationships between land values/housing prices and specific built environment factors. Recognizing the interconnected nature of built environment indicators is crucial, as not accounting for these interconnections may introduce bias in estimations. Therefore, it is essential to comprehensively explore various aspects of the built environment with land values. Additionally, studies using modern techniques such as machine learning mainly focus on improving the model's performance, lacking nonlinear and threshold effects of built environment features on land values.

3 Data and method

3.1. Study area and analytical framework

New York City is the economic centre of New York State, located in the northeastern United States of America (USA). New York City has five boroughs: Manhattan, Brooklyn, Queens, the Bronx, and Staten Island (Fig.1). In the heart of New York City, Manhattan is renowned for its iconic skyline, prestigious addresses, and exclusive luxury properties. It is home to some of the most expensive and sought-after residential neighbourhoods, such as Tribeca, Upper East Side, and Greenwich Village. The high land values in Manhattan are a testament to its status as a global financial and cultural hub, drawing affluent buyers and investors from across the globe. In the past, New York experienced a dramatic increase in land values. Between 2009 and 2014, NYC experienced a dramatic increase in land values, with the average land value per square foot rising by 50%, particularly in Manhattan, Brooklyn, and Queens. The substantial increase in land values posed challenges for urban planners. Therefore, it is essential to understand the underlying effects of the built environment features that usually cause buyers to be concerned.

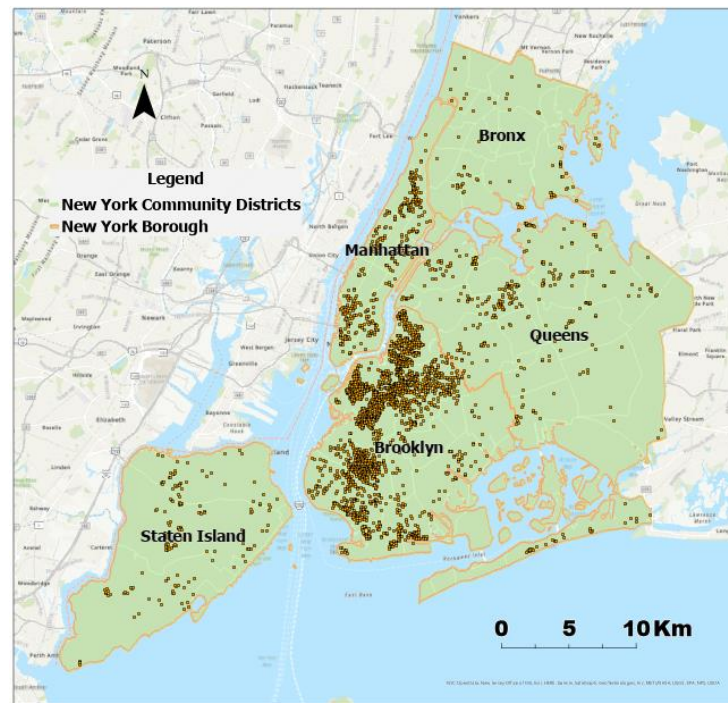


Figure 1 Study area

The research framework (Figure 2) comprises four major steps, beginning with research questions and finishing with the interpretation of the results. Firstly, two primary research questions guide this study: (1) understanding the significance of variables and (2) unravelling the nonlinear effects of the built environment on land values. Secondly, diverse data sources were collected for analysis. The land value data from MapPluto, published by New York Open Data, was utilized as the dependent variable. The independent variables included various aspects of the built environment. This

paper used the “5Ds” built environment approach to identify the impact of each built environment aspect on land values. The 5Ds built environment consists of density, diversity, design, destination of transit and destination accessibility. Thirdly, the data was divided into a training dataset (70%) and a validation dataset (30%). This study used 5-fold cross-validation and hyperparameter optimization to evaluate the model’s performance and address overfitting issues. GBDT and XGBoost models were utilized and compared, and the model with superior accuracy was chosen. However, machine learning models often pose interpretability challenges due to their black-box nature. Therefore, this study used Shapley to support model interpretation. Finally, the Shap values were employed to uncover the significant contributors and elucidate the nonlinear relationships.

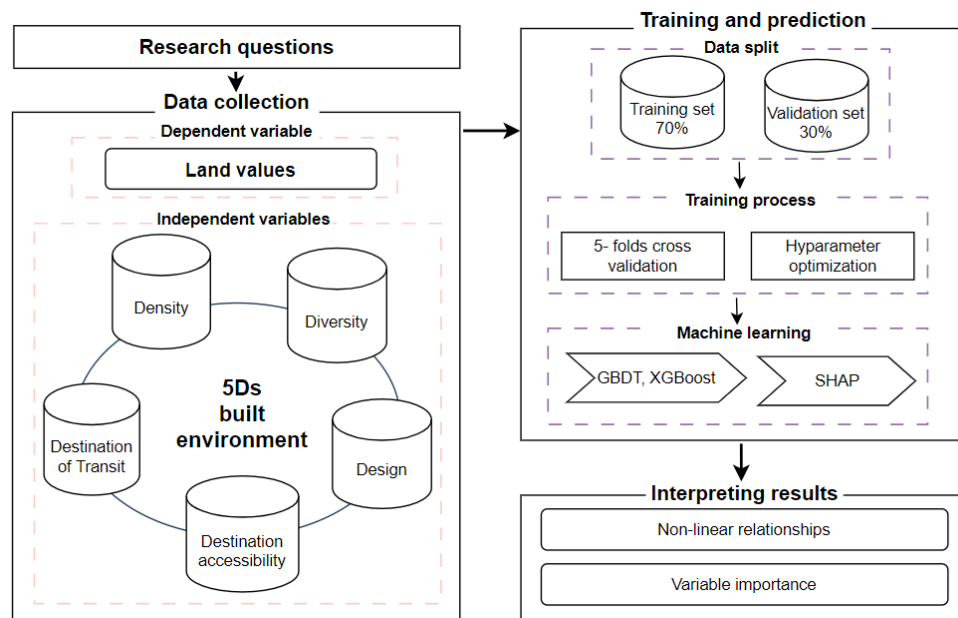


Figure 2 Analytical framework

3.2. Dataset

This study investigates the impact of built environment characteristics on land values. Two main types of data were used. The first data type used in this study is land values obtained from MapPLUTO. MapPLUTO encompasses diverse data types, such as land values, building information, and lot characteristics. In this study, we focused on residential lots to maintain a consistent dataset, reducing potential data discrepancies and enhancing the robustness of our results. Consequently, this study ignored commercial and entertainment lots, so only residential plots were analyzed. Following the filtration of residential lot conditions, 5048 lots were retained. There are two lot types, including walking building lots and elevator building lots. To avoid the significant gaps in land values between these lots, we excluded the elevator building lots and the remaining 4075 walking building lots. We used ‘price per square foot’ as the dependent variable, which is calculated by using the assessed total value divided by the lots’ area. The second data type pertains to the built environment and is

categorized into five dimensions, as previously explored in similar studies. This study considered five distinct built environment categories: density, diversity, design, destination of transit, and destination accessibility (Table 1).

The density category includes the built floor area ratio, resident floor area ratio, commercial floor area ratio and Facility floor area ratio as follows in studies of Deng et al. (2023), and Duan et al. (2024). The diversity category includes land use mix and POI diversity. The mixed land use and mixed POI are calculated using the entropy index as the following formula:

$$Mix = \frac{\sum_i^t W_i \cdot \ln(W_i)}{\ln(t)}$$

Where w_i is the proportion of the i^{th} land use purpose/ POI, t is the number of land use purposes/ POI type.

The design category emphasizes building and lot characteristics. The design variables used in this study include the number of buildings, number of floors, lot frontage, lot depth, building frontage, building depth, total building area, commercial area, residential area, office area, retail area, garage area, storage area, factory area, the number of condominiums, the relationship with other lots, construction year, lot's shape, relationship to neighbourhood buildings, and presence of building's basements.

The destination of transit illustrates transport accessibility. This research used the number of metro stations and bus stops within a radius of 400m from the land lots as indicators of accessibility. An increased number of transportation stations is associated with higher accessibility.

The destination accessibility category includes the number of each type of urban facility in a radius of 400m from the lots. Urban facilities used in this study include hospitals, schools, businesses, markets, theatres, parks, restaurants, banks, departments, convenience stores, Wi-Fi hotspots, universities, and galleries. Following previous studies, we included shorelines and the city centre accessibility in the analysis. The distance to the city centre is determined using network analysis with ArcGIS Pro. For shoreline, it was determined by considering plots within 500m of the coastline, with a value of 1 assigned to plots within this range and 0 otherwise. Finally, the location category illustrates the borough (Boro) where lots are located.

Table 1 Statistic of variables

	Variables	Mean	S.TD	Min	Max
Density	Floor area ratio	2.195	1.223	0.03	25.39
	Residential floor area ratio	2.158	1.189	0	10
	Commercial floor area ratio	0.171	0.871	0	10
	Facility floor area ratio	3.054	1.556	1	10
Diversity	Mixed land use	0.018	0.072	0	0.612
	Mixed POI	0.464	0.131	0	0.772
Design	Number of buildings	3.581	11.985	0	207

	Variables	Mean	S.TD	Min	Max
Destination of transit	Number of floors	3.523	1.588	0	32
	Lot frontage (feet)	41.361	84.823	0	3617
	Lot depth (feet)	92.112	108.142	0	5360
	Building frontage (feet)	25.938	66.296	0	3705
	Building depth (feet)	45.816	33.109	0	399
	Building area (square feet)	12800.39	22660.16	380	361955
	Commercial area (square feet)	144.911	1042.685	0	36036
	Residential area (square feet)	12666.85	22374.66	1000	361955
	Office area (square feet)	1.278	44.909	0	2430
	Retail area (square feet)	30.295	578.311	0	25689
	Garage area (square feet)	75.631	605.799	0	14076
	Storage area (square feet)	7.167	95.369	0	3762
	Factory area (square feet)	0.204	13.033	0	832
	Number of condominiums	2046.123	1578.67	1	5599
	Construction year (year)	50.768	46.727	2	231
	lot's shape (1: irregular shaped; 0: not irregular shaped)	0.155	0.362	0	1
	relationship to neighbourhood building 0: Not available; 1: Detached; 2: Semi-attached; 3: Attached)	0.513	1.053	0	3
	Building's basement	4.786	0.81	1	5
	Bus	5.12	4.21	0	26
	Metro	1.231	1.292	0	14
	N_school	2.56	3.071	0	20
	N_hospital	1.445	2.865	0	20
	N_business	57.539	60.272	0	517
	N_market	2.105	1.809	0	12
	N_theatre	0.412	1.132	0	15
	N_park	0.302	0.632	0	4
	N_restaurant	25.827	31.386	0	207
	N_bank	2.242	3.162	0	23
	N_department	0.234	0.719	0	7
	N_convenient	6.727	7.235	0	35
	N_wifi	7.316	11.853	0	87
	N_university	0.45	0.498	0	1
	N_gallery	1.685	7.643	0	151
	Shoreline (1: yes; 0: otherwise)	0.42	0.494	0	1
	Distance to Manhattan center (m)	8471.249	4785.684	175.27	31523.22
	Borough code (1: Manhattan; 2: Bronx; 3: Brooklyn; 4: Queens; 5: Staten Island)	2.937	0.82	1	5

3.3 Methodology

The primary objective of this study is to examine the built environment's impact on land values and analyze the presence of nonlinear relationships. Regarding nonlinear relationships, GBDT and XGBoost were frequently used to explore nonlinear relationships in previous studies. Therefore, GBDT and XGBoost are utilized in this study for land value prediction, and their performances are

rigorously compared. The model demonstrating superior performance is deemed more accurate, as gauged by accuracy indicators including R^2 , MAE, MSE, and RMSE.

3.3.1 Gradient Boosting Decision Tree

GBDT was used to investigate the nonlinear relationships between built environments and land values. By utilizing the GBDT method, the relative influence of each explanatory variable is revealed, enabling the identification of factors driving land values from different spatial scales and geographical contexts.

GBDT, a highly effective nonlinear modeling approach, was initially proposed by Friedman (2001). GBDT has several advantages that outweigh traditional regression models. Firstly, it accommodates continuous and categorical variables and exhibits robustness against outliers. As a result, GBDT proves to be a powerful technique suitable for regression and classification tasks. Secondly, GBDT demonstrates higher prediction accuracy and can handle missing data. Additionally, it enables examining intricate interactions among predictors and addresses multicollinearity (Dong et al., 2019). The essence of GBDT involves the iterative creation of subsequent generations of weak classifiers using base classifier classification regression trees, ultimately resulting in the globally optimal integrated artificial intelligence model. The GBDT model can be represented as follows:

$$F_m(x) = \sum_m^M w_m T(x; \theta_m)$$

Where w_m represents the weight of the tree with the optimal value being obtained through M iterations. $T(x; \theta_m)$ denotes the decision tree, where x represents the sample dataset; θ_m is the parameter set of the tree; and M is the number of trees. The loss function of the decision tree $T(x; \theta_m)$ was denoted as $L(y, F_m(x))$, is expressed as the squared error function. The current decision tree is indicated by $T_{m-1}(x)$ and GBDT determines the parameters of the subsequent decision tree by minimizing the lost function $\hat{\theta}_m$, which can be articulated as follows:

$$\hat{\theta}_m = \operatorname{argmin} \sum_{i=1}^N L(y_i; T_{m-1}(x_i) + T(x; \theta_m))$$

3.3.2 XGBoost

Extreme gradient boosting (XGBoost) is a powerful non-linear machine-learning technique suitable for both regression and classification tasks (Chen and Guestrin, 2016). XGBoost is an extension of gradient boosted decision trees that offers parallel tree boosting (Chen and Guestrin, 2016). It is an enhanced version of the popular Gradient Boosted Decision Tree (GBDT) algorithm, designed to overcome its computational limitations. However, XGBoost diverges from the GBDT approach by incorporating a second-order Taylor expansion in its loss function instead of the first-order Taylor

expansion employed by GBDT (Abbasniya et al., 2022). In XGBoost, a sequential ensemble approach, also known as sequential decision tree construction, is used to create a sequential Decision Tree (DT). Each sample in the dataset is assigned a weight that determines its likelihood of being selected for further analysis by a decision tree. At the outset, the weight assigned to each data point is initially uniform but undergoes variability due to statistical analysis (Bhati et al., 2021). XGBoost's objective function typically consists of 2 components: training loss and regularization, expressed as follows:

$$Obj(\theta) = L(\theta) + \Omega(\theta)$$

Where L represents the training loss function, which evaluates the model's performance on the training data. Ω denotes the regularization term, aimed at managing the complexity of the model such as overfitting. Furthermore, the complexity of each tree is calculated as the following:

$$\Omega(f) = \gamma T + \frac{1}{2} \cdot \lambda \sum_{j=1}^T \omega_j^2$$

Where T denotes the number of leaves and ω is the vector of scores assigned to the leaves. The structure score of XGBoost is defined as follows:

$$Obj = \sum_{j=1}^T \left[G_j \cdot w_j + \frac{1}{2} (H_j + \lambda) w_j^2 \right] + \gamma \cdot T$$

Where w_j are independent of each other. The form $G_j \cdot w_j + \frac{1}{2} (H_j + \lambda) w_j^2$ is quadratic and the optimal w_j for a given structure $q(x)$ is determined.

3.3.3 Shapley

Model interpretability in machine learning (ML) algorithms has been a significant concern that has often been insufficiently addressed. To tackle this issue, Lundberg and Lee (2017) introduced the SHAP (Shapley Additive Explanations) approach, which provides a method for interpreting ML models. While simple feature importance analysis can identify crucial components, its impact on prediction accuracy remains unclear. The primary advantage of SHAP interpretation lies in its ability to assess the influence of features on each sample, revealing both positive and negative effects (Kim and Kim, 2022). This approach facilitates a comprehensive understanding of models across various machine-learning frameworks. In SHAP, the contribution of each feature to the model output is determined based on its marginal contribution. Shapley values, which ensure fair allocation of feature contributions, are computed according to the following formula:

$$\phi_i = \sum_{S \in N} \frac{S! (n - |S| - 1)!}{n!} \cdot [v(S \cup \{i\}) - v(S)]$$

Where S refers to a subset of features used in the model, M denotes the number of features, and N represents the set of all input features. $[v(S \cup \{i\}) - v(S)]$ describes the deviation of Shapley values from their mean.

4 Results

4.1 Variable selection

Previous studies often implemented machine learning models without thorough testing of statistical significance. This practice can lead to selecting highly correlated variables, boosting model performance without capturing mechanistic effects from input to output. To address this, our approach begins with a multicollinearity test to identify potentially impactful variables. Variables with variance inflation factor (VIF) values exceeding 7.5 are systematically excluded (see Table 2). Subsequently, Ordinary Least Squares (OLS) regression is employed to assess the influence of explanatory variables on land values and selectively retain statistically significant variables. Notably, the model's performance improves by including these important variables. For instance, the R^2 increases by approximately 2.5% with only 19 variables, compared with 44 original variables. Following the previous studies, we extend our analysis using the Multiscale Geographically Weighted Regression (MGWR) method, revealing superior predictive performance over the OLS model. The MGWR model achieves a performance of 0.837, marking a 13.72% improvement over the OLS model (see Table 3). The spatial distribution of explanatory variables' coefficients is available in the Appendix. Notably, residual distribution tests indicate unsatisfaction with linear assumptions. Therefore, we used nonlinear machine learning models to explore the nonlinear relationships and threshold identification between the built environment and land values.

Table 2 Coefficients OLS model

Term	Coef	Std err	t	P> t
Const	4.1389	0.108	38.463	0.000***
Floor area ratio	0.2740	0.009	32.068	0.000***
Residential floor area ratio	0.1010	0.012	8.504	0.000***
Commercial floor area ratio	-0.0713	0.012	-5.765	0.000***
Mixed POI	0.4025	0.086	4.680	0.000***
Number of buildings	-0.0060	0.001	-6.969	0.000***
Number of floors	0.0888	0.006	13.658	0.000***
Lot Frontage	-0.0005	0.000	-4.391	0.000***
Building basement	0.0496	0.011	4.503	0.000***
N_bus	0.0062	0.003	2.229	0.026**
N_school	-0.0421	0.003	-12.127	0.000***
N_hospital	0.0184	0.004	4.784	0.000***
N_market	0.0138	0.006	2.236	0.025**
N_park	-0.0668	0.015	-4.370	0.000***

Term	Coef	Std err	t	P> t
N_restaurant	0.0049	0.001	8.812	0.000***
N_convenient	0.0061	0.002	3.549	0.000***
N_gallery	0.0108	0.005	2.229	0.026**
Shoreline	0.0799	0.002	3.989	0.000***
Distance to the city centre	-7.23e-05	2.97e-06	-24.346	0.000***
Borough	-0.1217	0.018	-6.778	0.000***
N	4075			
R ²	0.736			
Adj R ²	0.735			

Note: ***: Significant at 99% confidence level. **: Significant at 95% confidence level

Table 3 The performance comparison of OLS and MGWR

R ²			
Model	Selected 44 variables	Selected 19 variables	Improvement (%)
OLS	0.7181	0.736	+2.492
MGWR	-	0.837	+13.72% compared with the OLS

4.2 GBDT and XGBoost performances

This paper selects the most accurate model by comparing statistical indicators of machine learning algorithms, as shown in Table 4. Using 5-fold cross-validation and hyperparameter optimization, the optimal parameters were selected for land value prediction. Overall, the model's result showed good performance. GBDT showed a higher R² and lower MAE, MSE and RMSE than other models. The R² of the model is 0.840. GBDT, therefore, was selected as the most accurate model for land value prediction.

Table 4 Performance of GBDT and XGBoost model.

Model	MAE	MSE	RMSE	R ²
GBDT	0.320	0.194	0.441	0.840
XGBoost	0.322	0.197	0.444	0.838

4.3 Relative importance of variables

The study employed Gradient Boosting Decision Trees (GBDT) and SHAP analysis to assess the impact of built environment factors on land values. SHAP contribution analysis, depicted in Figure 3, highlights the importance of variables influencing land values, measured by their SHAP values. A higher Shap value signifies a more substantial contribution. The horizontal axis in Figure 3 represents the mean Shapley value of variables, indicating their average marginal contribution. The accompanying graph illustrates how each lot's variables contribute to its respective land values.

Figure 3-left illustrates the estimated Shap values of features ranked by important level. Figure 3 (left) ranks feature by their SHAP values, illustrating their importance. The top seven variables, in

decreasing order of contribution, were: built floor area ratio (29.18%), number of floors (18.14%), number of restaurants within 400m (12.62%), distance to the city center (10.25%), number of convenience stores, and number of galleries and boroughs. Surprisingly, the distance to the city center, often deemed crucial in previous studies, contributes only approximately 10.25% to land value prediction. Figure 3 - right depicts the local interpretation plot, where dots represent each land lot. The y-axis presents built environment features, and the x-axis illustrates the Shapley value. A positive association is observed between floor area ratio (Built_FAR), number of restaurants, galleries and land values. In contrast, the distance to the city centre is negatively associated with land values.

This part briefly presents the main findings: (1) The density aspect is the most important in land value prediction, with more than a third contribution; (2) destination accessibility is the second most important factor contributing to land values.

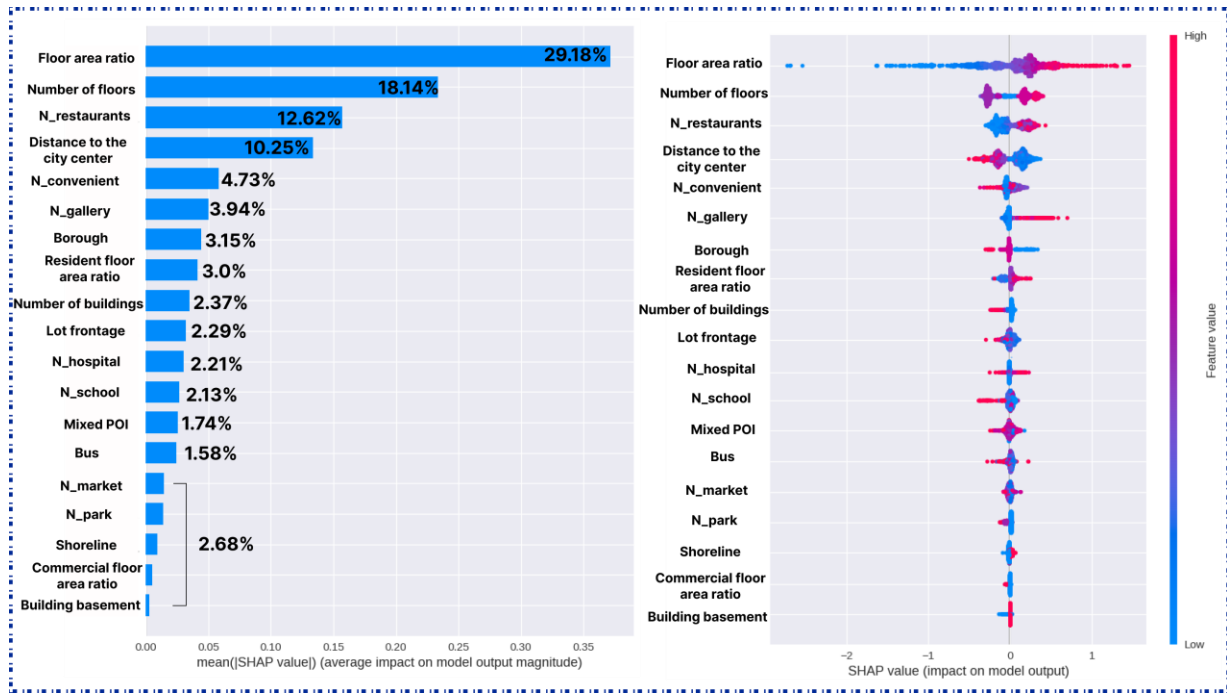


Figure 3 Relative importance of the built environment features

4.4 The non-linear relationship between the built environment and land value

While previous studies have acknowledged the significant contribution of the built environment to land values, much of the research has predominantly focused on linear relationships. This study, employing a more accurate machine learning model and SHAP, delves into the intricate micro-scale dynamics of how the built environment shapes land values. Key variables from each built environment dimension—density, diversity, design, transit destination, and accessibility—were selected to unravel non-linear relationships. These include density (floor area ratio, residential floor area ratio), diversity (Mixed POI), design (Number of floors, lot frontage, number of buildings), and

destination accessibility (N_restaurant, Distance to the city center, N_gallery, N_convenient). Notably, the transit destination's contribution was found to be minor and was therefore excluded from the analysis.

4.4.1 Density features

Figure 4 illustrates the relationship between density features and land values. Notably, the relationship between the built floor area ratio (Built_FAR), residential floor area ratio (Resid_FAR), and land values exhibits nonlinearity. Figure 4a shows a clear nonlinearity between floor area ratio and land value. Specifically, the Shapley value for Built_FAR increases rapidly for values below 6. Beyond a threshold of 6, the influence plateaus with the highest impact are observed in the 6-25 range. A similar trend is observed in the relationship between residential ratio and land value (Figure 4b). The influence on land values gradually increases before Resid_FAR reaches 4. However, beyond this threshold, the effect on land values shifts from positive to negative despite the lot being located near the city centre. Additionally, we observed an interaction between distance to the city centre and residential floor area ratio, with the highest SHAP values when Resid_FAR reaches ten and is near the city centre.

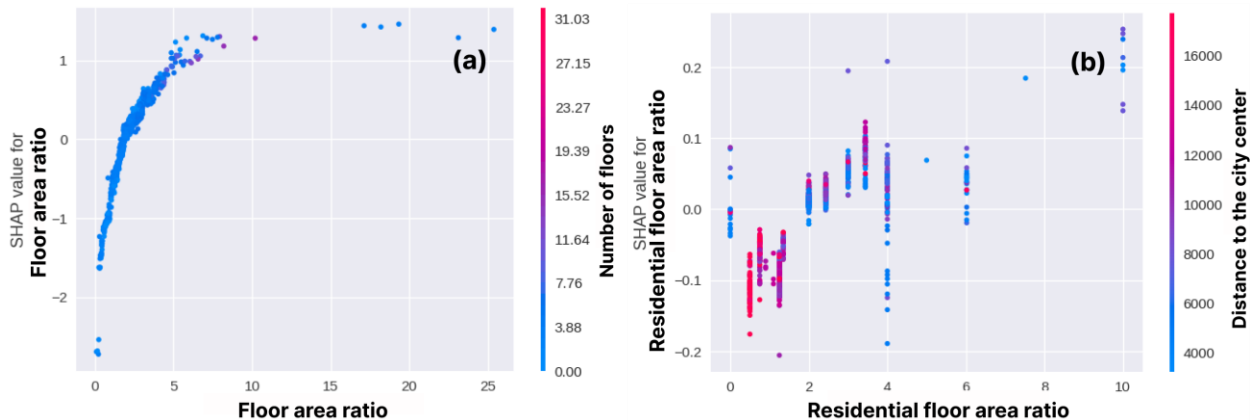


Figure 4 Non-linear relationship between density features and land values

4.4.2 Diversity features

Figure 5 illustrates the non-linear relationship between point of interest diversity and land values. The Shap values slightly decrease as mixed POI values range from 0.2 to 0.45. Upon exceeding the threshold of 0.45, the Shap value exhibits a significant increase, reaching a peak at mixed POI = 0.55 and gradually decreasing afterwards. This result indicates that the influence of mixed POI is highest if mixed POI achieves 0.55. Notably, there is a threshold at a mixed POI of 0.55, at which the impact of POI diversity on land values changes from positive to negative, indicating that crowded places and highly developed areas can have a reverse effect on land values.

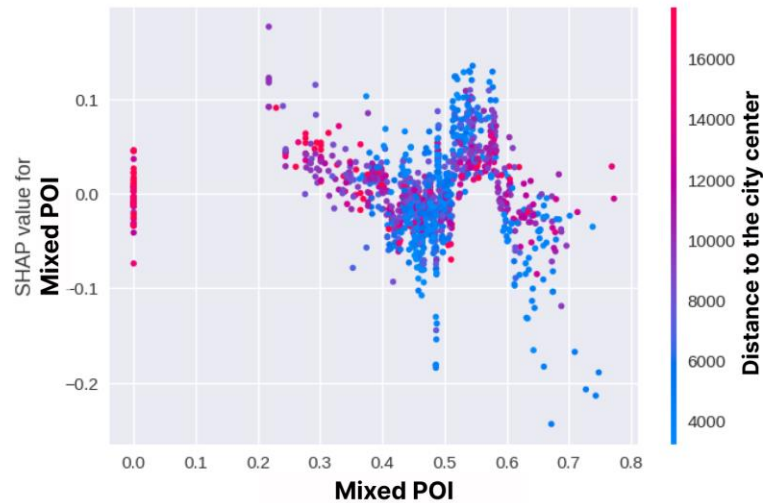


Figure 5 Non-linear relationship between diversity features and land values

4.4.3 Design features

Figure 6 presents the nonlinear relationship between design features and land values, focusing on the number of floors, buildings and lot frontage, which are recognized as influential determinants of land values among design attributes.

As depicted in Figure 6a, the effect on land value may be divided into two phases. Within the range of 0~5, the SHAP values exhibit a notable increase. However, as its range extends to 5~30, a slight decrement in the effect on land value is discerned, suggesting marginal shifts in impact. Lot frontage also has a nonlinear identification impact on land values. Specifically, beyond a threshold of 200, the effects of lot frontage on land values are minimal, as indicated by stable Shap values when lot frontage increases (figure 6b). Similarly, the number of buildings exhibits a nonlinear negative influence on land values, as illustrated in Figure 6c. Notably, we observed the interaction effects between the number of buildings within a radius of 400 feet and the floor area ratio. It is observed that lots featuring a higher number of buildings but with a lower floor area ratio (1.0~1.5) are found to exert a diminished impact on land values. The findings provided a new insight into the relationship between the number of buildings and land values, as most previous studies reported a positive association between building density and land values. This may be due to previous studies' lack of concern about the interaction between building density and floor area ratio. The high building density with a low floor area ratio may present less attractive places.

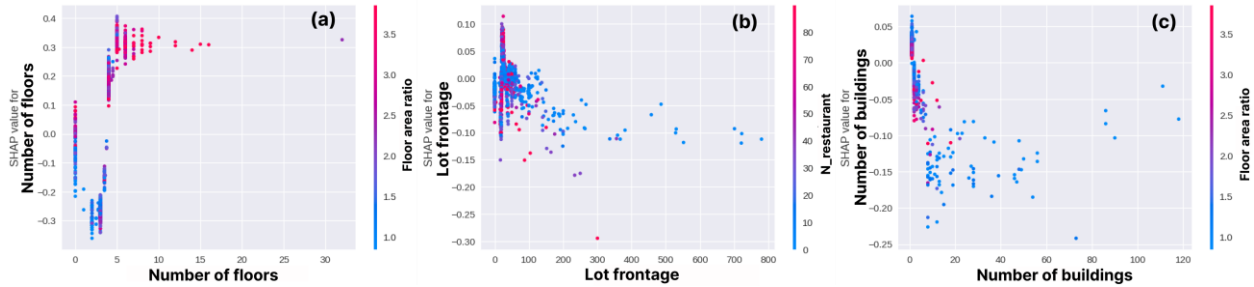


Figure 6 The non-linear relationship between design features and land values

4.4.4 Destination accessibility features

The nonlinear relationships between destination accessibility and land values are shown in Figure 7. Specifically, the number of restaurants, distance to the city centre, and galleries demonstrate nonlinear associations with land values.

Regarding the number of restaurants, the influence on land values increases significantly within the range of 0~50. When the number of restaurants exceeds 50 and below 90, the Shap value increases slightly before decreasing (Fig 7a). We observed that the impact of the number of restaurants on land value was the highest in the ranges of 50 to 90. The curve of the number of convenience stores reveals a nonlinear association (Fig 7b), which can be divided into two parts. First, the result revealed that the influence was positive for the first ten convenience stores but became negative as the number increased further. It is observed that the Shap value increases when the number of galleries is below a threshold of 12. However, exceeding a threshold of 12 results in a negligible increase in influence (Fig 7c). In contrast, the distance to the city centre exhibits a strong negative correlation with land values (Fig 7d). Initially, a slight decrease in influence is noted as the distance increases from 0 to 7 km. However, a significant decline is observed within the 7 to 10 km range. Beyond 10 km, the impact on land values diminishes gradually, reflected in decreasing SHAP values.

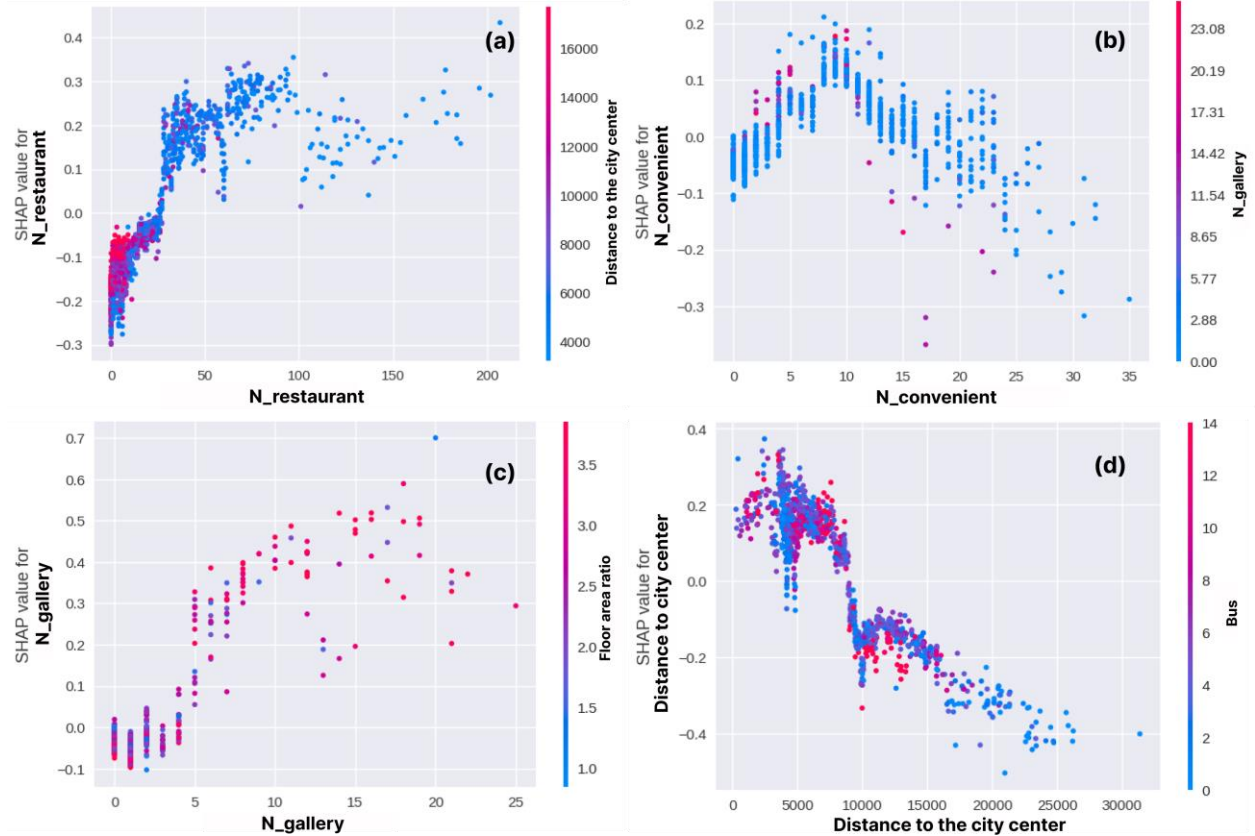


Figure 7 The non-linear relationship between destination accessibility features and land values

4.5. Interaction and local effects

Figure 8 depicts the interaction effects of the top 10 significant variables. Consisting of 100 sub-figures, it illustrates the interaction between each pair of selected features using SHAP interaction values. In each sub-figure, the x-axis represents the SHAP interaction value, distinguishing negative effects with negative values and positive effects with positive values. The colour gradient, transitioning from blue to red, indicates the feature values ranging from low to high. Each column corresponds to the SHAP interaction values of the feature specified by the column label. The figure includes a diagonal sub-figure, representing the main effect of the feature (first-order interaction), along with ten off-diagonal sub-figures, elucidating the higher-order interactions between features.

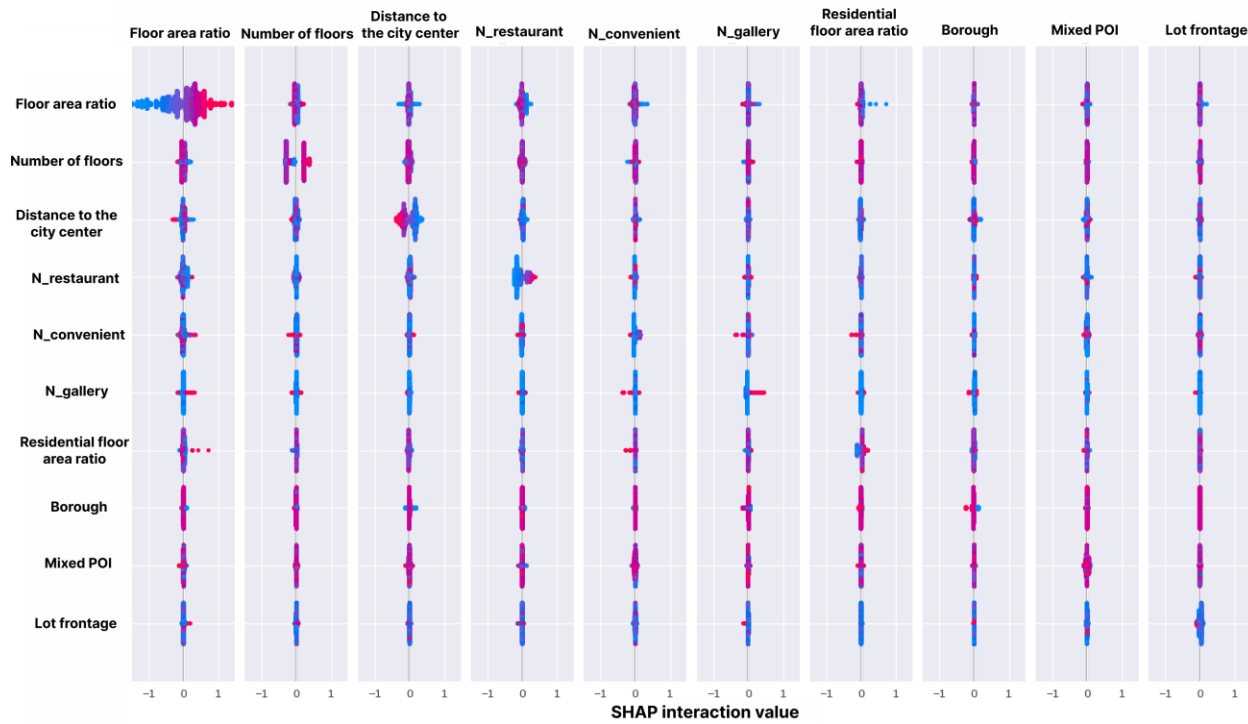


Figure 8 SHAP interaction values among explanatory variables

To illustrate the influence of the built environment on land values, this study employed local interpretation to measure the contribution of different features to the land value prediction. Unlike conventional methods based on linear assumptions that only provide global feature importance, SHAP analysis offers a more granular understanding by quantifying the contribution of each observation. Figure 8 presents the effects of features on land values as determined by the GBDT model trained on the dataset. In the figure, the red color indicates features that positively impact land values, while the blue color represents features with a negative effect. In this analysis, five lots were randomly selected. The first lot's Shapley value is 3.45 with significant contributions: floor area ratio and distance to the city center. The built environment variables in the red zone, including the residential floor area ratio and borough code, indicate that those variables contribute to an increase in land values. This is reasonable, given that the first lot is located far from the city center and has limited urban facilities. In contrast, most variables positively contribute to the third case, located in the center of the Manhattan district, where the built environment has been developed over many years. For instance, with a Shapley value of 7.02, this lot is positively influenced by the floor area ratio, the number of restaurants, the number of floors, and distance to the city center. Notably, most built environment variables negatively impact the land value of the fifth lot, with the most significant variable situated in the blue zone.



Figure 9 The local explanations for random cases

5 Discussion

This work explores the roles played by the built environment and land values. It contributes to the state of the art by uncovering the non-linear relationship of the built environment features. Several key findings are discussed in this section.

Our findings partially align with existing studies, underscoring the substantial impact of density features on land value. Density, specifically the built floor area ratio, emerges as a pivotal factor. Notably, a rapid increase in land values accompanies an elevation in the built floor area ratio, contradicting the findings of Fesselmeyer and Seah (2018), who reported a negative influence of localized density on property values. It is crucial to highlight that the influence on land values does not continuously increase beyond a built floor area ratio of 10. These findings contribute to a nuanced understanding of the intricate dynamics identified in previous research on this topic (Gao et al., 2006, He et al., 2010). Our results also indicate the interaction between floor area ratio and the number of buildings, which aligns with the study by Fesselmeyer and Seah (2018). For instance, the number of buildings within 400m decreases the impact on land values for plots with a low floor area ratio. This suggests a buyer preference for locations with a high floor area ratio and low building density. Chen et al. (2023) and Glaeser et al. (2005) revealed that floor area ratio and population density positively correlate with land value. Our general findings are consistent with this proposition, suggesting that the residential floor area ratio positively impacts land value prediction. Chen et al. (2023) and Zhou et al. (2020), however, do not account for non-linear relationships and interactions. This study further investigated the relationship between floor area ratio and land values across varying distances from the city center.

Secondly, this study investigates the significance of destination accessibility as the second most influential factor shaping land values for walking building lots. Consistent with previous studies

(Tam et al., 2022, Yuan et al., 2022), the findings highlight a negative relationship between distance to the city center and land values, contributing approximately 8.5% to land values. Notably, the relationship is non-linear, as the influence on land values decreases slightly within a distance range of 0 to 7 km, experiences a significant drop between 7 to 10 km, and stabilizes beyond 10 km. Moreover, the number of restaurants, galleries, and businesses positively associated with land values, aligning with prior research (Abidoye and Chan, 2017, Hu et al., 2020). In contrast to the frequently emphasized impact of destination accessibility on land values, we found it has a relatively weak effect on land values if the number of destinations exceeds thresholds, supported by some recent studies such as Tang et al. (2018) and Cellmer (2023), who emphasizes both positive and negative influences of specific housing-related categories on prices. These findings underscore that urban facilities may have adverse effects on land values.

Thirdly, this study emphasizes the importance of building design on land values, particularly the number of floors, which contributes 5.73% to land values. This finding is consistent with previous studies on houses' technical conditions (Jandásková et al., 2022). Notably, the contribution of design features to land values follows non-linear patterns. Specifically, the impact of the number of floors in land values gradually increases when the number of floors is below 5 but slightly decreases when the number exceeds 5. Compared with the literature, the relatively smaller importance of the lot's frontage is consistent with Doan (2023). However, we argue the mechanism behind the effect of the lot's frontage is different. While Doan (2023) indicated that the buyers prefer small frontage because of affordable prices, we hypothesize that the moderate effect of street vibrancy on the relationship between land values and frontage can explain the case in New York. Small block sizes and frontage can encourage people to walk and participate in street activities and create a high street vibrancy level, which might increase high land values. This study also highlighted non-linear and threshold effects, whereas other studies used linear assumptions.

Additionally, our findings indicate that the presence of metros and bus stations has a minor contribution to land values. This result contrasts with studies by Malaitham et al. (2020), Zhao and Ke (2021), which suggest a significant influence of metro systems on property prices and land values. However, our results are supported by Zhou et al. (2021), who found that while a new subway might lead to a slight increase in property prices, its impact on land values remains minor. This phenomenon can be attributed to the fact that public transit tends to significantly affect land values in low-income areas and regions lacking adequate transit infrastructure, while its impact is relatively insignificant in high-income areas with well-developed transit services, such as New York City (Chen et al., 2023).

Furthermore, our study sheds light on the relationship between diversity and land values. We observe a positive association between mixed POI and land values, with the mixed POI contributing approximately 1.74% to land value estimation, in line with findings by Li et al. (2021). However, due to the lack of threshold identification, previous research has overlooked the potential negative impact

of excessive POI diversity on land values. Our study addresses this gap by examining how diversity impacts land values across different ranges of mixed POI. We find that an excessively high mixed POI index can negatively affect land values, a finding consistent with Wang et al. (2022). Residents are likely to prefer areas with appropriate points of interest and avoid those with extremely high POI diversity, which can bring noise, chaos, and pollution, leading to safety concerns (Song and Knaap, 2004). These factors may reduce neighborhood attractiveness, negatively impacting land values (Matthews and Turnbull, 2007, Abdullahi et al., 2015).

6 Conclusion

Previous studies on the relationships between the built environment and land values have limitations related to generalization, nonlinear relationships, and the identification of thresholds. Focusing on data from New York City, this study employed nonlinear machine learning and SHAP algorithms to explore the threshold impacts of the 5D built environment on land values. Two machine learning techniques were employed, with the results indicating the superior performance of GBDT over XGBoost, OLS, and MGWR in land value estimation. The study confirmed the nonlinear relationship of density, diversity, design, destination of transit, and destination accessibility on land values using the GBDT model and SHAP method. The study illustrates that density and destination accessibility are the most significant contributors, accounting for 32.18% and 31.94% of land value variation, respectively. The top three factors affecting land values were built floor area ratio, number of floors, and number of restaurants. The study also suggests threshold relationships for built environment features that affect land values, offering valuable insights for urban planners and house price management authorities. The contributions of this study are as follows:

- (1) Uncovering the relative importance of the built environment in land value prediction using interpreted machine learning algorithms and SHAP.
- (2) Revealing the nonlinear relationships between the built environment and land values helps explain the mixed results in previous studies.
- (3) This study highlights the threshold effects of built environment features on land values, which were overlooked in the literature.

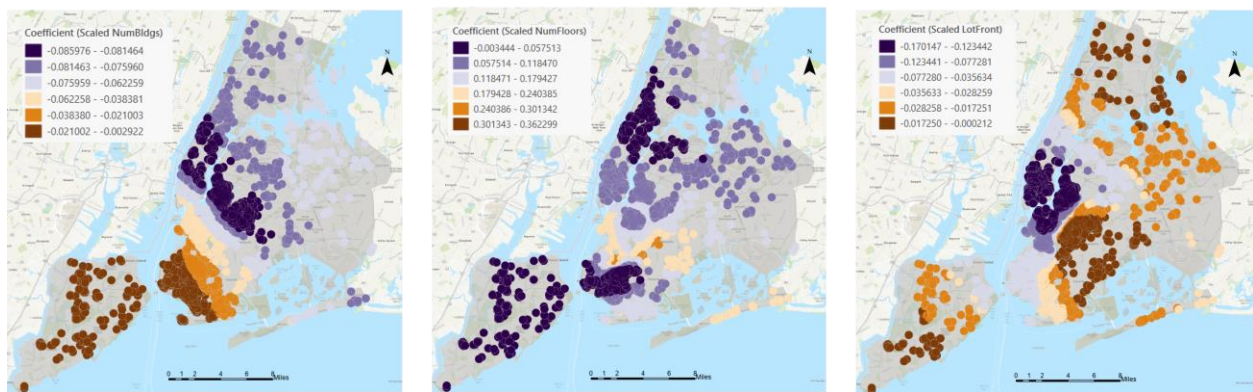
The knowledge derived from this study can serve as a reference for policymakers and managers. Planners and policymakers can leverage these findings to develop flexible zoning regulations that consider the varying impacts of different built environment features on land values. For instance, areas with higher levels of density or a diverse range of amenities may be designated for specific types of development to maximize land values. A better understanding of the nonlinear relationship can inform investment decisions by identifying areas where the threshold of built environment features is positively associated with higher land values. This information can be strategically used to invest in properties that align with the preferences of potential buyers or tenants.

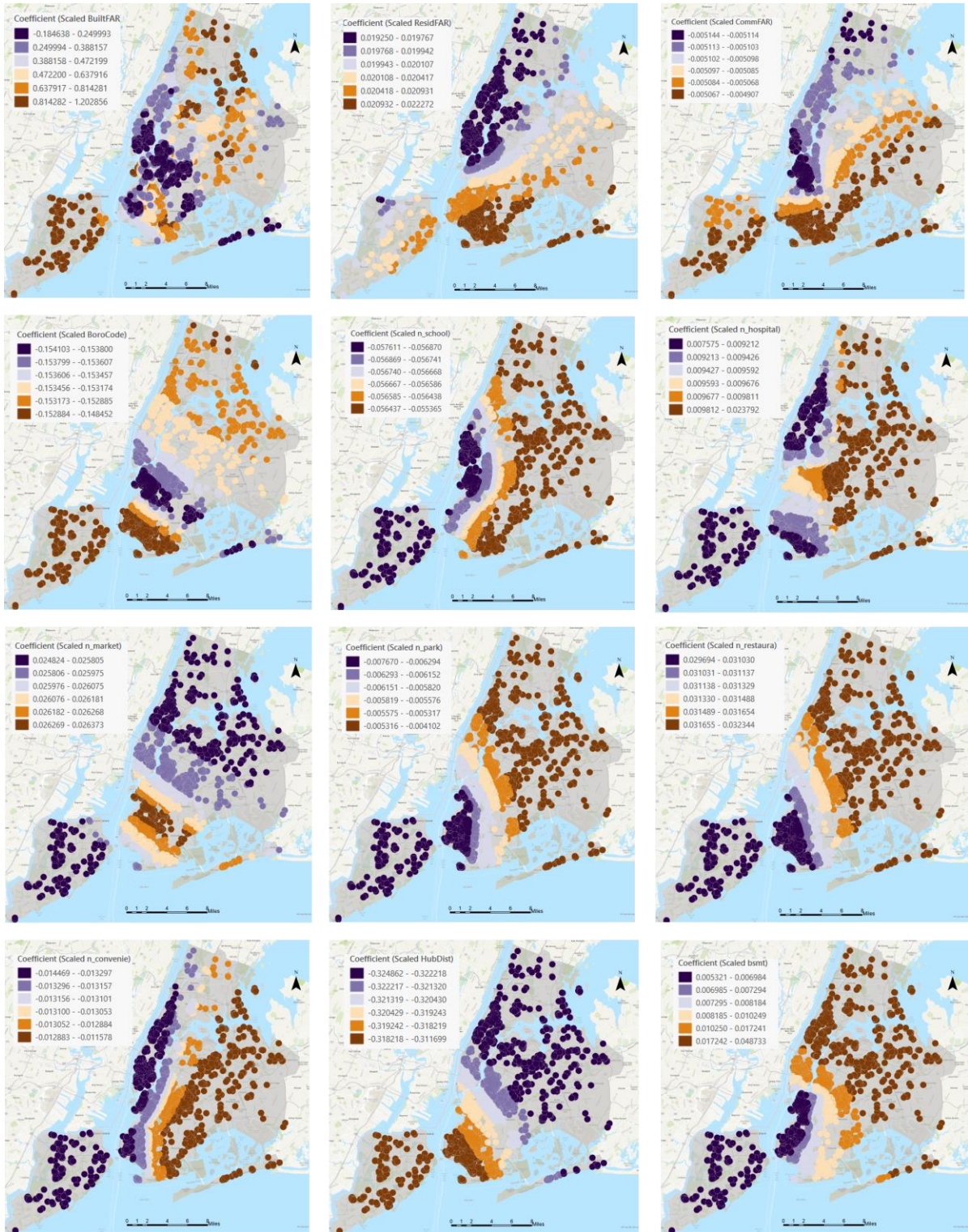
Furthermore, policymakers can develop recommendations to support sustainable urban development. These policies may include incentives for developers to incorporate certain built environment features that positively influence land values or tax policies that encourage property owners to invest in improving their properties' environmental quality.

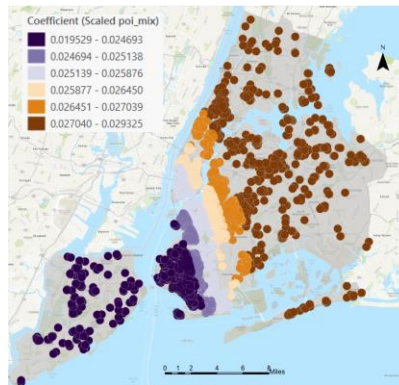
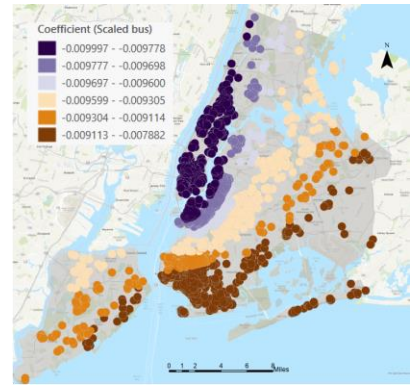
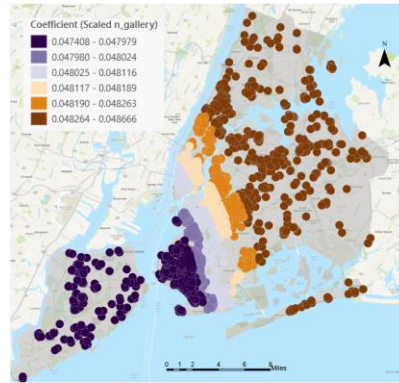
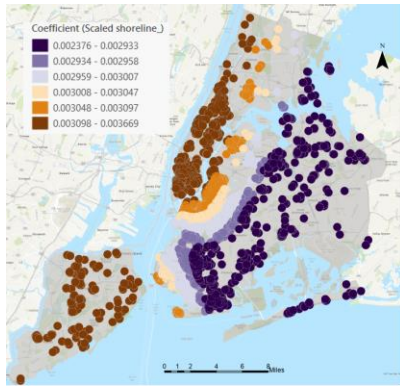
Notwithstanding the groundbreaking discovery of the link between land prices and the built environment, a few disadvantages still need to be considered. First, using borough codes of five districts is the only way we have control over the potential impact of socioeconomic features on land values. Second, the land value data for a single year is employed in this study. By using more data and conducting a more in-depth analysis, findings for different cities across multiple years might be verified.

Appendix: MGWR result

MGWR results







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